Pre-analysis plan: Testing daughters, gender attitudes and political preferences in the SOEP data.

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1. Sample and inclusion criteria.

In this study, we will test whether the gender attitudes and political preferences of individuals in the general population are affected by whether they have daughters. We will use data from the Socio-Economic Panel (SOEP). The SOEP is a representative, multi-cohort survey that has been running since 1984. Every year, individuals in households throughout Germany are surveyed by a survey institute on behalf of DIW Berlin. These respondents provide information on topics such as their income, employment history, education, and health. Because the same people are surveyed every year, it is possible to track long-term psychological, economic, societal, and social developments. To keep pace with changes in society, additional samples are added regularly, and the survey is adapted accordingly. For the objective of this study, we will focus on the period 2000 to 2020.

For each hypothesis, we will include respondents with 1. complete data on the dependent variable which is relevant for that hypothesis, 2. with complete data on the independent variables and the controls that are used in each specification, 3. who have at least one child defined as their "Son/daughter" and less than nineteen children and 4. for whom there is information about the gender of all children defined as "Son/daughter".

2. Survey questions and variables.

We will use data on the following survey questions in our analyses. The phrasing is based on source questionnaire files posted on the SOEP webpage.

2.1. Survey questions on the number and gender of the children in the household.

The SOEP uses interviewers, and the interviewer's collect information about all members of the respondent's household, including gender (male, female) and year of birth. We can obtain the total number of biological and adoptive kids for each adult (question: "Do you have or had children? If so, how much?", variable *sumkids*) as well as the age and sex of each child. For each adult, we will include data about the number of children and the gender of the children for up to 19 children, as that is the maximum in the SOEP data. Only individuals with at least one child reported will be included in the analysis samples. The variable codes for the child gender in the SOEP data are *"kidsex01"* to *"kidsex19"* for the first to nineteenth child.

We will create a dummy variable called *d_daughters* that assigns 1 to individuals who have at least one daughter and 0 otherwise and nineteen dummy variables indicating the number of children (a dummy variable that takes value 1 if the individual has two children and 0 otherwise, a dummy

variable that takes value 1 if the individual has three children and 0 otherwise, and so on and so forth until 19 children).

2.2. Gender attitudes.

We will include the following questions (A-G) about gender attitudes to test whether having daughters affects the opinions toward female-male roles. Questions A and B are on the 1 (totally disagree) to 7 (completely agree) agree-disagree scale, while questions C to G are on the 1 (totally disagree) to 5 (completely agree) agree-disagree scale. We specify the code that identifies the question in the SOEP database in parentheses:

I will read you a series of statements. To what degree do you personally agree with each statement?

A: … Children below the age of 6 suffer if their mother works (plh0298_v1/v2)

B: … Children below the age of 3 suffer if their mother works (plh0302_v1/v2)

C: … It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally (plh0308_v1/v2)

D: … Men should be involved in housework (plh0303)

E: … Children suffer when father focuses on his professional career (plh0304)

F: … Men should care more about financial security (plh0307)

G: … Working mothers can establish just as warm and secure a relationship with their children as mothers who do not work (plh0309)

Questions A-C are included in the 2012 and 2018 waves, while questions D-G are only included in the 2012 wave. We will use PCA to build a summary variable of all the answers to questions A-G and denote this variable *gender_attitudes* (this variable will be only defined for the 2012 wave).

2.3. Survey questions on political preferences.

We will include the following question (H) about general political preferences on a 0-10 left (0) to right (10) scale:

H: "In politics, people often talk about "left" and "right" when describing different political views. When you think about your own political view, how would you rate them on the scale below?" (plh0004)

This question is included in 2005, 2009, 2014 and 2019. We will store the answer to question H in a variable that we denote *political_preferences.*

2.4. Survey questions on time use and behaviors.

We will include the following questions (I-N) about time use and behaviors to test the time that fathers and mothers spend on household work and on children care:

How many hours do you spend on:

I: … Childcare on Saturday? (pli0019)

J: … Childcare on Sunday? (pli0022)

K: … Childcare on weekdays? (pli0044)

L: … Chores on Saturday? (pli0012)

M: … Chores on Sunday? (pli0016)

N: … Chores on weekdays? (pli0043)

These questions are included in 2001, 2003, 2005, 2007, 2009, 2011, 2012, 2013, 2015, 2017, and 2019. We only include observations that report data for all the previous variables. We will construct the dependent variable measuring gender-equal behaviors by summing the number of hours reported in all categories I–N for each individual, divided by the sum of the hours reported in categories I–N by himself/herself and his/her opposite sex partner. We will denote this variable *housework* (thus, for individuals who do not live with an opposite sex partner, these dependent variable will not be defined).

2.5. Background characteristics.

In the analysis we will control for a range of individual-level characteristics. The specifications will include information about the gender (*sex*) and age (*gebjahr*) of the father/mother. Additionally, we include controls for the parent's region of birth - whether they were born in former East or West Germany - (*sampreg*), migration background (*migback*), and the education level measured by years of education (*pgbilzeit*) of the father/mother will be added. We refer to this set of variables as the *vector of controls* and assume that they are predetermined.

We also incorporate fixed effects for the survey round in specifications that comprise more than one round along with the set of dummy variables for the number of children (with 1 child being the baseline category).

3. Hypothesis and tests.

All the hypothesis will be tested using OLS regressions, with standard errors clustered at the individual level. We will interpret a two-sided p-value <0.05 in the significance test of the coefficient of interest as statistically significant evidence.

First hypothesis: Having daughters leads to stronger support for gender equality (as measured by attitudes).

- *Specification 1:* The dependent variable will be *gender_attitudes* (see section 2.4.). The independent variable will be *d_daughters*.

Second hypothesis: Having daughters leads to more left-leaning political preferences.

- *Specification 2:* The dependent variable will be *political_preferences* (see section 2.1.). The independent variable will be *d_daughters*.

Third hypothesis: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

- *Specification 3:* The dependent variable will be *housework* (see section 2.3.). The independent variable will be *d_daughters*.

Fourth hypothesis: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

- *Specification 4*: Same as Specification 1 of the first hypothesis, but adding an interaction term between gender and the independent variable.

This regression specification follows approaches commonly used in literature. Yang et al. (2025), for instance, employed two models to examine the impact of having daughters on political preferences. In their first specification, the key independent variable was the number of daughters, complemented by a gender dummy and a set of individual-level controls. In the second specification, they replaced the main variable with a dummy indicator for having at least one daughter. Similarly, Cronqvist and Yu (2017) used a daughter dummy to investigate whether U.S. companies led by CEOs with daughters exhibit higher corporate social responsibility ratings. A notable distinction in their approach is that they did not control for the number of children, included CEOs without children in the analysis, and accounted for firm- and CEO-level characteristics in some specifications. Glynn and Sen (2015) also used a daughter dummy in their regressions, controlling for the number of children in select analyses.

Finally, we follow Stöcker et al. (2024) and Hamjediers (2021) in selecting and testing variables to measure gender attitudes. Both studies use questions related to gender attitudes, but they chose to create a simple mean variable and use it as the dependent variable. In contrast, we decided to run a PCA to construct our dependent variable.

References.

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